

TASK-1

Title: RTL Design & Functional Simulation (Verilog)

Objective: Design and verify an 8-bit ALU using Verilog HDL and Vivado Simulator.

Tools Used: Vivado Simulator, Verilog HDL

alu_8bit.v

RTL code:

```
module alu_8bit(  
    input [7:0] A,  
    input [7:0] B,  
    input [3:0] sel,  
  
    output reg [7:0] Y,  
    output reg carry,  
    output reg zero,  
    output reg overflow  
);  
  
reg [8:0] temp;  
  
always @(*) begin  
  
    Y = 8'b00000000;  
    carry = 0;  
    overflow = 0;
```

```
case(sel)
```

```
// ADD
```

```
4'b0000: begin
```

```
    temp = A + B;
```

```
    Y = temp[7:0];
```

```
    carry = temp[8];
```

```
    overflow = (~(A[7]^B[7])) & (A[7]^Y[7]);
```

```
end
```

```
// SUB
```

```
4'b0001: begin
```

```
    temp = A - B;
```

```
    Y = temp[7:0];
```

```
    carry = temp[8];
```

```
    overflow = ((A[7]^B[7]) & (A[7]^Y[7]));
```

```
end
```

```
// AND
```

```
4'b0010: begin
```

```
    Y = A & B;
```

end

// OR

4'b0011: begin

Y = A | B;

end

// XOR

4'b0100: begin

Y = A ^ B;

end

// NAND

4'b0101: begin

Y = ~(A & B);

end

// NOR

4'b0110: begin

Y = ~(A | B);

end

// SHIFT LEFT

4'b0111: begin

```
Y = A << 1;
```

```
end
```

```
// SHIFT RIGHT
```

```
4'b1000: begin
```

```
Y = A >> 1;
```

```
end
```

```
default: begin
```

```
Y = 8'b00000000;
```

```
end
```

```
endcase
```

```
// ZERO FLAG
```

```
if(Y == 8'b00000000)
```

```
zero = 1;
```

```
else
```

```
zero = 0;
```

```
end
```

```
endmodule
```

tb_alu_8bit.v

Testbench Code:

```
module tb_alu_8bit;
```

```
    reg [7:0] A;
```

```
    reg [7:0] B;
```

```
    reg [3:0] sel;
```

```
    wire [7:0] Y;
```

```
    wire carry;
```

```
    wire zero;
```

```
    wire overflow;
```

```
    integer i;
```

```
    alu_8bit DUT(
```

```
        .A(A),
```

```
        .B(B),
```

```
        .sel(sel),
```

```
        .Y(Y),
```

```
        .carry(carry),
```

```
        .zero(zero),
```

```
        .overflow(overflow)
```

```
    );
```

initial begin

// ADD

A = 8'd20;

B = 8'd10;

sel = 4'b0000;

#10;

// SUB

A = 8'd50;

B = 8'd25;

sel = 4'b0001;

#10;

// AND

A = 8'b11001100;

B = 8'b10101010;

sel = 4'b0010;

#10;

// OR

sel = 4'b0011;

#10;

```
// XOR
```

```
sel = 4'b0100;
```

```
#10;
```

```
// NAND
```

```
sel = 4'b0101;
```

```
#10;
```

```
// NOR
```

```
sel = 4'b0110;
```

```
#10;
```

```
// SHIFT LEFT
```

```
A = 8'b00001111;
```

```
sel = 4'b0111;
```

```
#10;
```

```
// SHIFT RIGHT
```

```
sel = 4'b1000;
```

```
#10;
```

```
// RANDOM TESTS
```

```
for(i = 0; i < 20; i = i + 1)
```

begin

A = \$random;

B = \$random;

sel = \$random % 9;

#10;

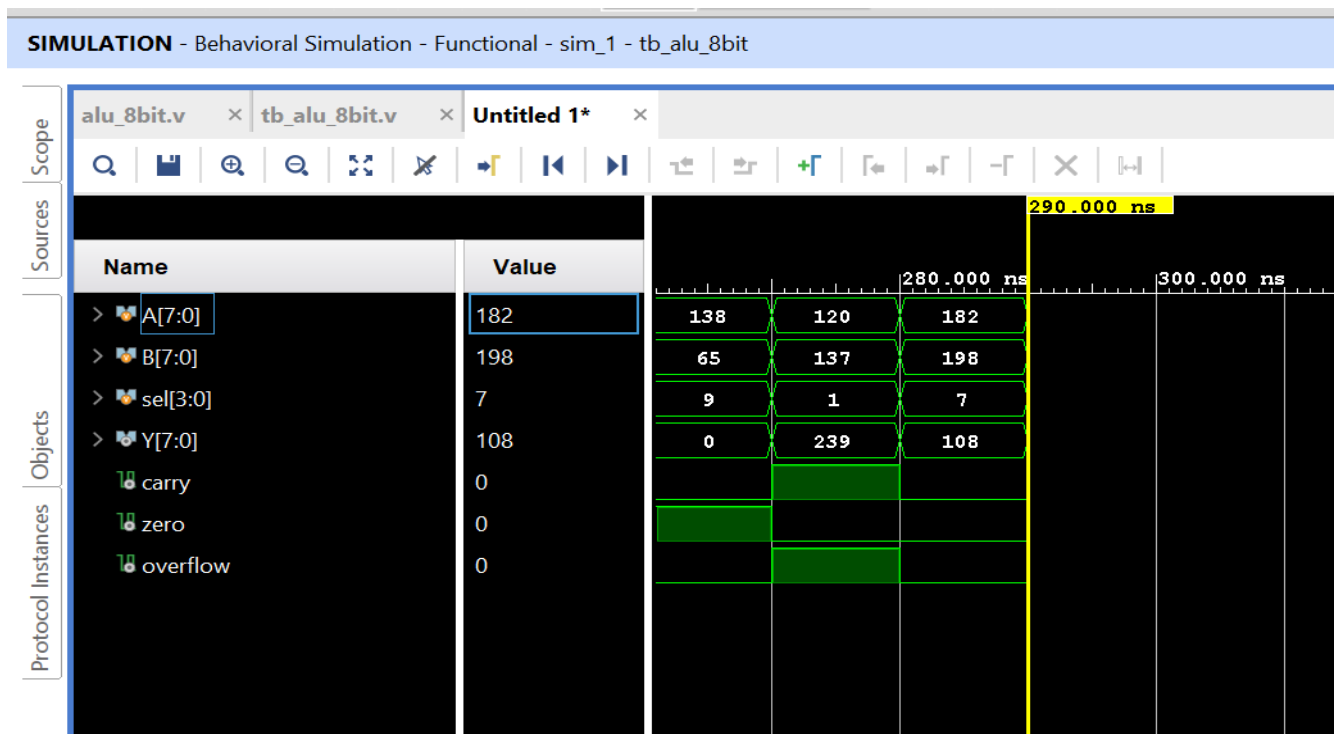
end

\$finish;

End

endmodule

Waveform Screenshot:



Explanation:

- An 8-bit ALU was designed using Verilog HDL and verified using Vivado Simulator.
- The ALU supports arithmetic, logical, and shift operations including addition, subtraction, AND, OR, XOR, NAND, NOR, left shift, and right shift.
- Status flags such as carry, zero, and overflow were implemented.
- Verification was performed using directed and random test cases. Simulation waveforms confirmed correct functionality for all operations.

Conclusion: The RTL design and functional verification of the 8-bit ALU were successfully completed. Simulation results confirmed correct operation for all test cases.